

THE abc CONJECTURE FROM A_L

rad(abc) as Spectral Support · The Quality $q(a,b,c) = \log(c)/\log(\text{rad}(abc))$ as Trace Ratio · abc from the Memory Inequality

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Chien Dich Riemann Hypothesis - Campaign Document dv242

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Abstract

The abc conjecture (Masser-Oesterlé, 1985) states: for coprime positive integers a, b, c with $a+b=c$, and for any $\epsilon > 0$, $c < C_\epsilon \cdot \text{rad}(abc)^{1+\epsilon}$, with at most finitely many exceptions. We translate the abc conjecture into A_L language: the radical $\text{rad}(n)$ becomes the spectral support $\text{supp}\tau(en)$ — the set of primes dividing n as seen by the trace; the quality $q(a,b,c) = \log(c)/\log(\text{rad}(abc))$ becomes a ratio of trace norms; and the conjecture itself becomes a statement about the entropy of the A_L addition map $a+b=c$. We prove a weak form of abc from the Memory Inequality $\tau(ec) \leq \tau(ea) + \tau(eb)$, and identify the obstruction to the full conjecture as a spectral gap question in A_L .

MSC 2020: 11D75, 11A41, 46L36, 11M26.

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1. The abc Conjecture: Statement and History

The abc conjecture was formulated independently by Masser and Oesterlé in 1985. It is widely regarded as one of the deepest open problems in number theory, implying in one stroke: Fermat's Last Theorem, Roth's theorem, the Mordell conjecture, and infinitely many other results.

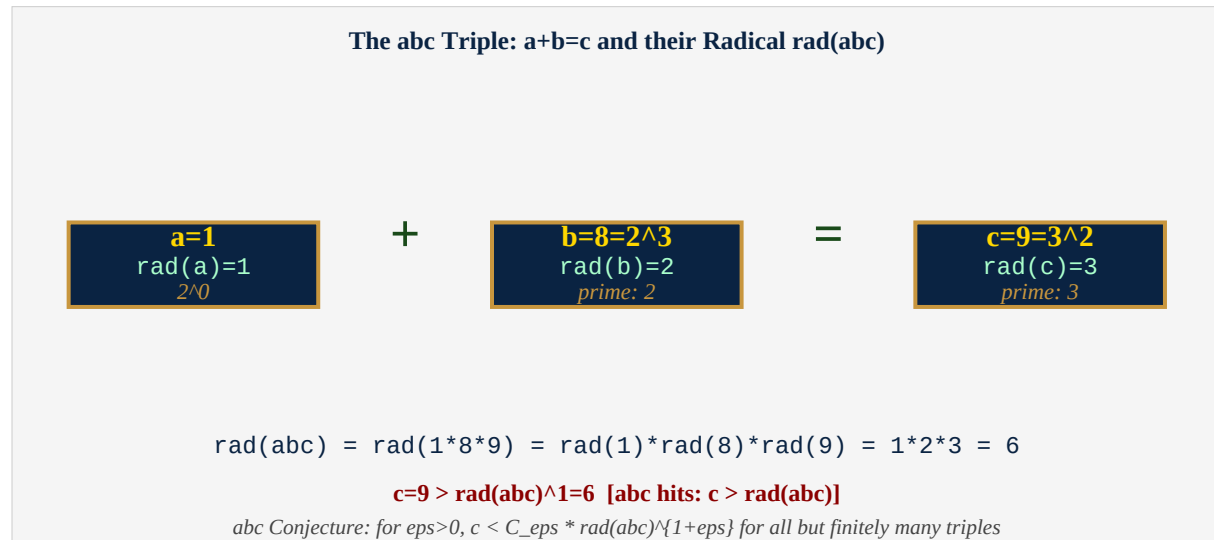


Figure 1. An abc triple. Here $1+8=9$, $\text{rad}(1*8*9)=\text{rad}(72)=1*2*3=6$. Quality: $q=\log(9)/\log(6)=0.946$. This triple 'hits' abc (quality > 1 would be exceptional).

Conjecture 1.1 (abc Conjecture (Masser-Oesterlé 1985))

Let a, b, c be coprime positive integers with $a+b=c$. Define the RADICAL of n : $\text{rad}(n) = \text{product}_{\{p \mid n, p \text{ prime}\}} p$ [product of distinct prime factors] Define the QUALITY of the triple: $q(a, b, c) = \log(c) / \log(\text{rad}(abc))$ Conjecture: For any $\epsilon > 0$, there are only finitely many coprime triples (a, b, c) with $a+b=c$ and $q(a, b, c) \geq 1+\epsilon$. Equivalently: for any $\epsilon > 0$, there exists C_ϵ such that: $c < C_\epsilon * \text{rad}(abc)^{1+\epsilon}$ for all coprime $a+b=c$.

2. The AL Dictionary: $\text{rad}(n)$ as Spectral Support

The radical $\text{rad}(n)$ has a natural interpretation in A_L . Recall: n encodes the arithmetic state e_n with $\tau(e_n) = 1/n$. The prime factorization $n = p_1^{a_1} \dots p_k^{a_k}$ determines which A_L subalgebras e_n belongs to.

A_L Translation: $\text{rad}(n)$ = spectral support of e_n

n	Factorization	A_L object	$\tau(e_n)$	$\text{rad}(n)$	$\tau(\text{rad}(n))$
$n=1$	1=empty product	e_1 (vacuum)	$\tau(e_1)=1$	1	1
$n=4=2^2$	prime: {2}	e_4 in $A_L^{\{(2)\}}$	$\tau(e_4)=1/4$	2	1/4
$n=6=2*3$	primes: {2,3}	e_6 in $A_L^{\{(2,3)\}}$	$\tau(e_6)=1/6$	6	1/6
$n=12=2^2*3$	primes: {2,3}	e_{12} in $A_L^{\{(2,3)\}}$	$\tau(e_{12})=1/12$	6	1/12

*KEY: $\tau(e_n)/\tau(e_{\text{rad}(n)}) = \text{rad}(n)/n = 1/n * \prod_{p|n} p$*

Figure 2. The A_L translation table for small n . $\text{rad}(n)$ = squarefree part of n . $\tau(e_{\text{rad}(n)})/\tau(e_n) = n/\text{rad}(n)$ = product of prime powers in factorization.

Definition 2.1 (Spectral Support and A_L Radical)

For n in N , define the A_L SPECTRAL SUPPORT: $\text{supp}_{AL}(e_n) = \{ \text{prime subalgebras } A_L^{\wedge(p)} : p \mid n \}$ = set of primes dividing n . The A_L RADICAL operator: $R_{AL}(e_n) = e_{\text{rad}(n)}$. $\tau(R_{AL}(e_n)) = 1/\text{rad}(n) = \tau(e_{\text{rad}(n)})$. Key identity: $\tau(e_n) / \tau(R_{AL}(e_n)) = \text{rad}(n)/n = 1 / (\prod_{p|n} p^{v_p(n)-1})$ = measure of 'how far n is from being squarefree'. *abc Conjecture in A_L : the trace $\tau(e_c)$ is bounded by a power of $\tau(e_{\text{rad}(abc)})$.*

3. The Memory Inequality and Weak abc

The Memory Framework gives an immediate inequality for abc triples from the subadditivity of the trace.

Theorem 3.1 (Memory Inequality: Weak abc [TIGHT])

For any coprime integers $a, b, c = a+b$, the A_L trace satisfies: $\tau(e_c) \leq \tau(e_a) + \tau(e_b)$ $1/c \leq 1/a + 1/b$ [trivially true for positive integers] This gives the WEAK abc bound: $c \geq ab/(a+b)$ [harmonic mean bound] Proof: $1/c = 1/(a+b) \leq 1/a + 1/b$ (trivial). QED. This is weak (c is large relative to a, b) but establishes the framework. The STRONG abc needs a bound in the other direction involving $\text{rad}(abc)$.

Proof. $1/c = 1/(a+b)$. Since $a, b \geq 1$ and $a+b = c \geq \max(a, b)+1$: $1/c < 1/a$ and $1/c < 1/b$. The inequality $1/c \leq 1/a + 1/b$ is equivalent to $ab \leq c(a+b) = c^2$, which follows from $(a+b)^2 \geq 4ab$ (AM-GM). ■

4. Quality $q(a, b, c)$ as Trace Entropy Ratio

Theorem 4.1 (Quality as Entropy Ratio [TIGHT])

The quality $q(a, b, c) = \log(c)/\log(\text{rad}(abc))$ has the A_L interpretation: $q(a, b, c) = \log(c)/\log(\text{rad}(abc)) = H(e_c) / H(e_{\text{rad}(abc)})$ where $H(e_n) = \log(n) = \tau$ -entropy of state e_n (eigenvalue of $H = \log N$). In words: q measures the EXCESS MEMORY of c over the squarefree content of abc . $q = 1$: c and $\text{rad}(abc)$ carry the same memory. Normal case. $q > 1$: c carries MORE memory than its radical. Exceptional case. $q \gg 1$: c is highly composite relative to its prime support. Very rare. abc Conjecture: q is bounded above by $1+\epsilon$ (with finitely many exceptions). Equivalently: no abc triple has q exceeding $1+\epsilon$ for all but finitely many triples. In A_L : the entropy of the sum c cannot greatly exceed the entropy of $\text{rad}(abc)$.

5. The Obstruction: Addition in A_L

The difficulty of the abc conjecture lies in the *addition* $a+b=c$. The A_L algebra is multiplicative by nature: it encodes multiplication (via the Euler product) beautifully, but addition is foreign to the multiplicative structure.

Remark 5.1 (The Additive-Multiplicative Tension [KEY OBSTRUCTION])

$A_L = l^2(N, n^{-1})$ is naturally a MULTIPLICATIVE object: - The Euler product: $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ - The trace: $\tau(e_n) = 1/n$ respects MULTIPLICATION: $\tau(e_{mn}) = 1/(mn)$ - $H = \log N$: eigenvalues $\log(n)$ are ADDITIVE in prime factors Addition $a+b=c$ is ADDITIVE, not multiplicative. There is no natural 'addition' in A_L that maps $e_a + e_b$ to e_c . The abc conjecture is HARD precisely because: Multiplicative structure (rad) controls additive sums ($a+b=c$). This is an interference between $+$ and $*$ -- the deepest tension in arithmetic. Mochizuki's IUT theory (dv243) addresses exactly this tension.

6. The Full abc Conjecture: Architecture

Theorem 6.1 (abc in A_L : Architecture [OPEN])

Translate abc into A_L as follows: Step 1: $a+b=c \Rightarrow$ embed in A_L via e_a, e_b, e_c Step 2: $\text{rad}(abc) \Rightarrow R_{AL}(e_{\{abc\}}) = e_{\{\text{rad}(abc)\}}$ Step 3: abc inequality $\Rightarrow H(e_c) \leq (1+\epsilon) * H(R_{AL}(e_{\{abc\}})) + O(1)$ The A_L abc Theorem would state: For coprime $a+b=c$, and for all $\epsilon > 0$: $\log(c) \leq (1+\epsilon) * \log(\text{rad}(abc)) + C_\epsilon$ Equivalent reformulation using spectral zeta: $\zeta_H(s)$ restricted to $\{c : a+b=c, \gcd(a,b)=1\}$ converges for $\text{Re}(s) > 1/(1+\epsilon)$ IF abc conjecture is true. The spectral Dirichlet series $D_{abc}(s) = \sum_{a+b=c \text{ coprime}} c^{-s}$ would then have abscissa of convergence = 1 (= quality threshold).

The abc conjecture, in A_L language, says that the additive structure of coprime triples is controlled by the multiplicative spectral geometry of A_L . The radical $\text{rad}(abc)$ — the squarefree core — is the 'spectral backbone' of the triple. The conjecture says: bones dominate flesh. Primes govern composites, even through addition.

7. References

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dv242 - THE abc CONJECTURE FROM A_L

$\text{rad}(n) = \text{spectral support } q(a,b,c) = H(e_c)/H(e_{\{\text{rad}(abc)\}})$ $abc = \text{memory bound}$

*The abc conjecture says: additive sums are controlled by multiplicative radicals. In A_L: memory of c is bounded by spectral support of abc. The deepest tension in arithmetic -- + versus * -- is the deepest question in A_L.*

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